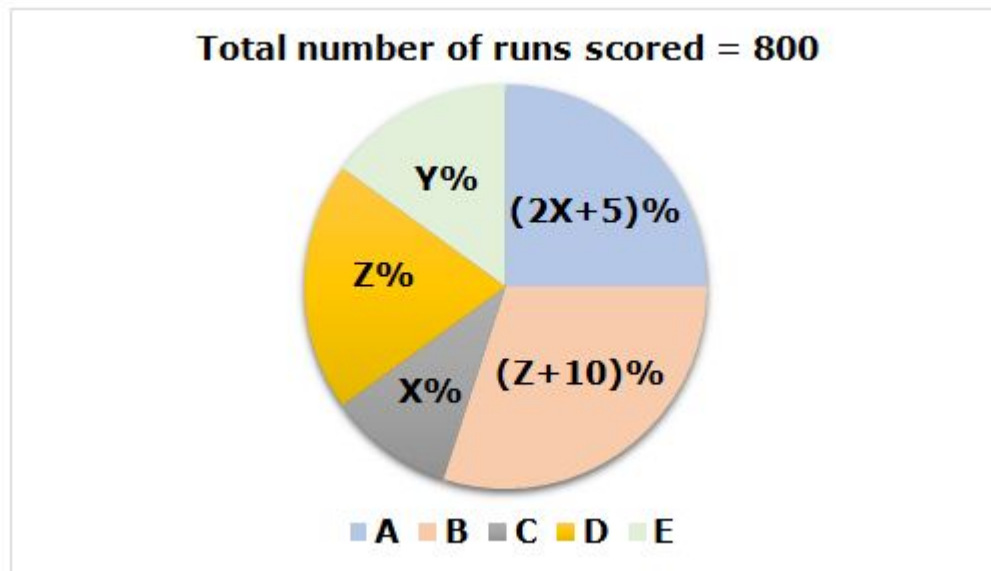


1. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of runs scored by five batters: A, B, C, D, and E.



Note:

The strike-rate = $\left[\left\{ \frac{\text{the number of runs scored}}{\text{the number of balls faced}} \right\} \times 100 \right]$

The number of balls faced by A, B, C, D, and E is 160, 160, 100, 320, and 100 respectively.

The ratio of the strike rate of A and B is 5:6.

The difference between the strike rate of C and D is 30. ($C > D$)

If $\frac{1}{4}^{\text{th}}$ of the total number of runs scored by A is scored in boundaries (fours + sixes) and the ratio of the number of fours and sixes scored by him is 8:3, then find the strike rate of runs scored in boundaries by A.

Note: Strike rate of boundaries = $\left(\frac{\text{Runs scored from boundaries}}{\text{Total balls faced in which boundaries were hit}} \right) \times 100$

- a. 366.67
- b. 454.54
- c. 233.33
- d. Cannot be determined
- e. None of these

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2. Questions

If the number of dot balls faced by C is [N], then his strike rate, excluding dot balls faced by him, will be increased by 25%.

Which of the following is/are definitely true about [N]?

I: Exactly divisible by 5 and 15.

II: Odd integer.

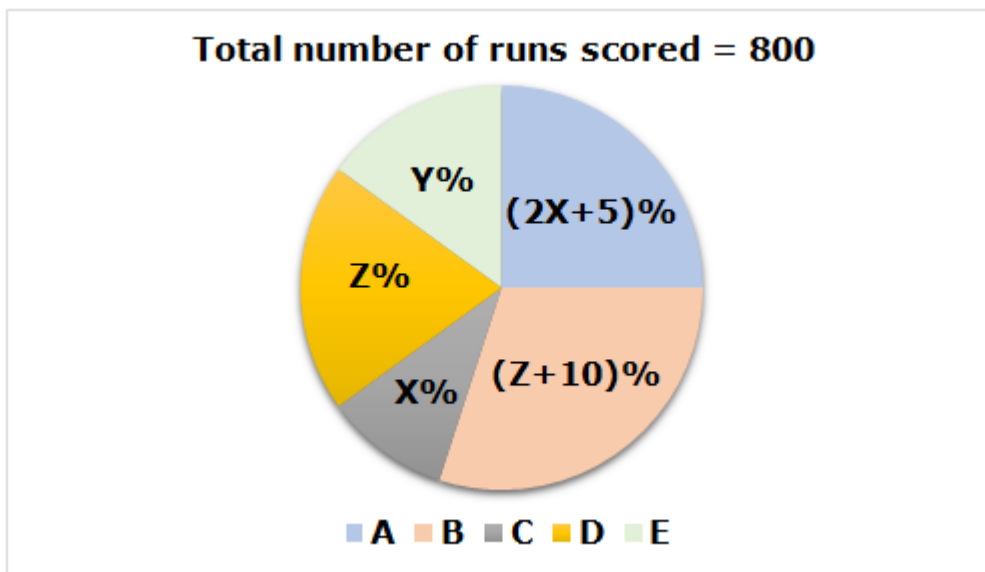
III: Even composite number.

- a. Only I
- b. Both I and II
- c. Only III
- d. All I, II and III
- e. None is true

3. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of runs scored by five batters: A, B, C, D, and E.



Note:

The strike-rate = $\left[\left\{ \frac{\text{the number of runs scored}}{\text{the number of balls faced}} \right\} \times 100 \right]$

The number of balls faced by A, B, C, D, and E is 160, 160, 100, 320, and 100 respectively.

The ratio of the strike rate of A and B is 5:6.

The difference between the strike rate of C and D is 30. ($C > D$)

If there is another batter F, who faced 10 balls less than the number of balls faced by B, and the strike rate of F is 100 less than the number of runs scored by him, then find the ratio of the number of runs scored by E and F.

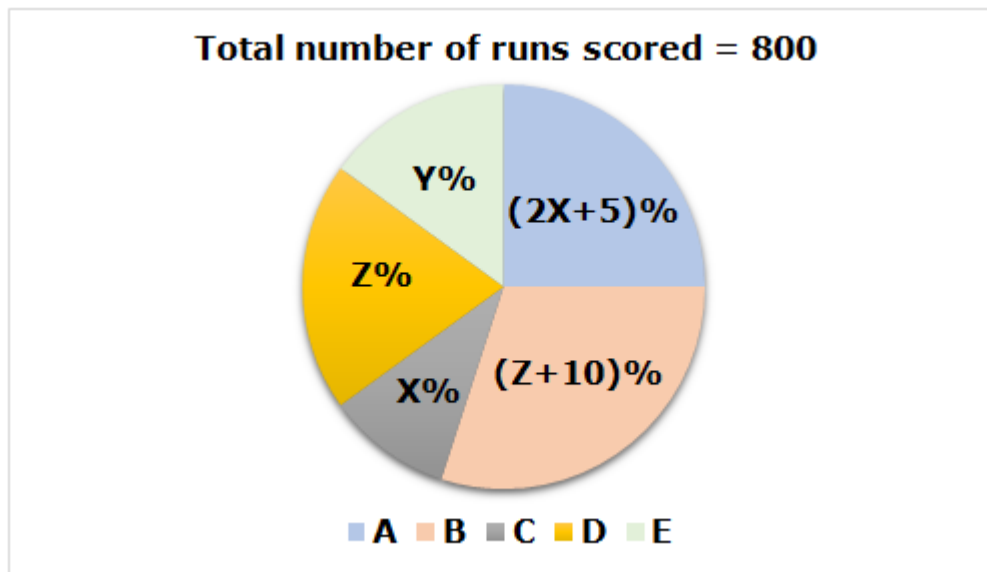
- a. 2:5

- b. 1:3
- c. 3:5
- d. 1:2
- e. None of these

4. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of runs scored by five batters: A, B, C, D, and E.



Note:

The strike-rate = $\left[\left\{ \frac{\text{the number of runs scored}}{\text{the number of balls faced}} \right\} \times 100 \right]$

The number of balls faced by A, B, C, D, and E is 160, 160, 100, 320, and 100 respectively.

The ratio of the strike rate of A and B is 5:6.

The difference between the strike rate of C and D is 30. ($C > D$)

If the number of runs scored by D is increased by [P]% and the number of balls faced by him is reduced by [Q], then his strike rate will be 100. ($Q - P = 30$)

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Determine the correct relation between [P] and [Q].

- a. $P > Q$
- b. $P < Q$
- c. $P \geq Q$
- d. $P \leq Q$
- e. $P = Q$

5. Questions

Which of the following is/are correct?

I: $[X+Z] = 2Y$

II: $[Z-Y] = [X/2]$

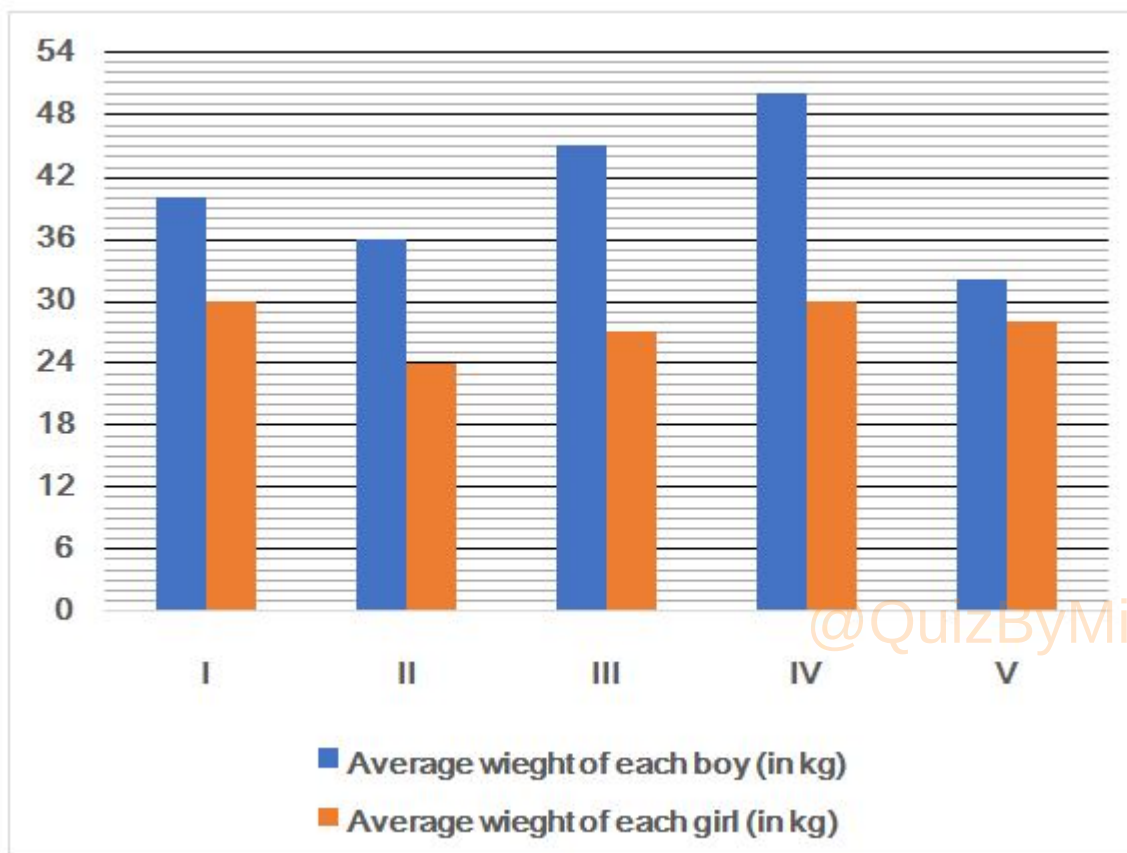
III: $[(X*Y)/10] = [Z-5]$

- a. Only I follow
- b. Both I and II follow
- c. Only III follow
- d. All I, II and III
- e. None follow

6. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the average weight (in kg) of each boy and each girl in five different classes I, II, III, IV and V of a school.



Note:

The total number of students in the class = the number of boys in the class + the number of girls in the class

The total number of students in class I, II, III, IV, and V is 200, 120, 360, 100, and 400 respectively.

The total weight of all students (boys + girls) in classes I, II, III, IV, and V is 7200 kg, 3840 kg, 13320 kg, 4500 kg, and 12000 kg respectively.

In class I, the number of boys who belong to the rural area is equal to the number of girls who belong to the urban area. If the ratio of the number of boys who belong to the urban area and girls who belong to the rural area is 9:5, find the total number of girls who belong to the rural area.

- a. 30
- b. 40
- c. 60
- d. 50
- e. 20

7. Questions

Out of the total students in class II, 62.50% participated in sports and the rest participated in arts. If the number of boys who participated in arts is 30, then find the ratio of the number of girls who participated in sports and arts from class II respectively. (Each student participated in either sports or arts but not both)

- a. 5:3
- b. 3:2
- c. 4:1
- d. Cannot be determined
- e. None of these

8. Questions

In class III, if the number of boys who left the school is [X] and the number of girls who joined the school is [Y], then the total number of students will be reduced by 10 and the new ratio of the number of boys and girl will be 1:1. Find the value of [Y].

- a. 33
- b. 26
- c. 10
- d. 25
- e. 15

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9. Questions

The total number of students in another class VI is $\frac{3}{4}$ th as that of class V and the total weight of all

students (boys + girls) together in class VI is 8100 kg. If the average weight of each boy and each girl in class VI is 30 kg and 21 kg respectively, then find the difference between the number of boys and girls in class VI.

- a. 100
- b. 150
- c. 120
- d. 180
- e. None of these

10. Questions

In class IV, the average age of each boy is 25% more than the average age of each girl in the same class. If the average age of all students (boys + girls) in class IV is 9.50 years, then find the total age of all boys in class IV.

- a. 200 years
- b. 750 years
- c. 425 years
- d. 575 years
- e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the percentage distribution of the total number of cakes (chocolate + vanilla + strawberry) sold by five shops P, Q, R, S and T. It also shows the number of chocolate cakes sold and the ratio of the number of vanilla and strawberry cakes sold by each shop.

Shop	The percentage of the total number of cakes sold (chocolate + vanilla + strawberry)	The number of chocolate cakes sold	The ratio of the number of vanilla and strawberry cakes sold
P	24%	120	2:1
Q	16%	64	3:1
R	32%	240	3:2
S	20%	120	1:1
T	8%	56	5:3

Notes:

The total number of cakes sold = the number of chocolate cakes sold + the number of vanilla cakes sold + the number of strawberry cakes sold

The number of vanilla cakes sold by shop P is 56 more than the number of strawberry cakes sold by shop Q.

If the total number of cakes unsold by shop S is 25% less than shop R, then the ratio of the total number of cakes made (sold + unsold) by shop R and S will be 20:13.

I = The percentage of the total number of cakes sold by shop R, out of the total number of cakes made (sold + unsold) by the same shop.

II = The difference between the total number of cakes sold and unsold by shop S.

- a. $I > II$
- b. $I < II$
- c. $I \geq II$
- d. $I \leq II$
- e. $I = II$

12. Questions

In shop P, if $\frac{3}{4}$ th of the total number of cakes (chocolate + vanilla + strawberry) sold are sold to males. If the ratio of the number of chocolate, vanilla, and strawberry cakes sold to females is 3:2:1, then find what percent of the total number of cakes sold to males are vanilla?

- a. 75%
- b. 60%
- c. 80%
- d. Cannot be determined
- e. None of these

13. Questions

Which of the following is/are definitely true?

I: The number of vanilla cakes sold by shop Q is 10% less than shop P.

II: The ratio of the number of chocolate and strawberry cakes sold by shop S is 3:1.

III: The difference between the total number of cakes sold by shops R and T is 200.

- a. Both I and III
- b. Only II
- c. Both I and II
- d. All I, II and III
- e. None is true

14. Questions

The total number of cakes sold by another shop U is 20 less than shop S. In shop U, if the difference between the number of chocolate and vanilla cakes sold is equal to the number of strawberry cakes sold, which is equal to 20, then find the number of chocolate cakes sold by shop U. (chocolate cakes > vanilla cakes)

- a. 30
- b. 80
- c. 40
- d. 60
- e. 90

15. Questions

If the ratio of the price of each vanilla cake in shop Q and T is 3:5 and the difference between total revenue generated from the sale of all vanilla cakes by shop Q and T respectively is Rs. 1410, then find the difference between the price of each vanilla cake in shops Q and T.

- a. Rs. 24
- b. Rs. 28
- c. Rs. 22
- d. Rs. 20
- e. None of these

16. Questions

Read the following information carefully and answer the questions based on it.

Certain number of people (males + females) have two types of accounts i.e., savings account and current account in two banks X and Y.

Bank X: The total number of people who have account (savings + current) is 2000, out of which 60% are savings account. The ratio of the number of males who have savings account and the number of females who have current account is 3:1. The number of females who have savings account is $\frac{3}{5}$ th of the number of males who have current account.

Bank Y: The total number of people (males + females) who have saving account is 1000, which is twice of the total number of people (males + females) who have current account. The number of females who have current account is 16.67% of the number of males who have savings account. The number of females who have savings account is equal to the number of males who have current account.

Notes:

The total number of people who have account = the number of people who have savings account + the number of people who have current account

The number of people who have savings account = the number of males who have savings account + the number of females who have savings account

The number of people who have current account = the number of males who have current account + the number of females who have current account

Each person has one account.

Each person has either savings or current account, but not both.

In bank X, $\frac{5}{7}$ th of the total number of males who have accounts (savings + current) have active account. If the ratio of the number of males whose savings account is active and dormant is 2:1, then find what percent of the number of males whose current account is dormant?

- a. 12%
- b. 20%
- c. 35%
- d. Cannot be determined
- e. None of these

17. Questions

In bank Y, if 25% of the total number of males who have savings account are senior citizen, which is 70 more than the number of female senior citizens who have savings account, then find the total number of non-senior citizens (males + females) who have savings account.

- a. 420
- b. 330
- c. 770
- d. 640
- e. None of these

18. Questions

The number of males who have current account in bank Y is what percent of the total number of people who have account (savings + current) in bank Z?

Statement I: 75% of the total number of people who have accounts in bank Z are savings account holders.

Statement II: The difference between the number of people who have savings and current account in bank Z is 1200.

- a. Statement I alone is sufficient
- b. Statement II alone is sufficient
- c. Either statement I or II alone is sufficient

- d. Both statements I and II together are necessary
- e. Neither statement I nor II is sufficient

19. Questions

Quantity I: The difference between the total number of males and females who have account (savings + current) in bank X.

Quantity II: The total number of females who have account (savings + current) in bank Y.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

20. Questions

In bank X, if the ratio of the number of females who have savings and current account becomes 5:3, while the number of females who have current account remained unchanged, then find the percentage increase in the total number of females who have account (savings + current) in bank X.

- a. 33.33%
- b. 37.50%
- c. 62.50%
- d. 66.67%
- e. None of these

Explanations:

1. Questions

The number of runs scored by A = $(2X+5)\%$ of 800 = $(16X+40)$

The number of balls faced by A = 160

The strike-rate of A = $[\{(16X+40)/160\} \times 100] = (10X+25)$

The number of runs scored by B = $(Z+10)\%$ of 800 = $(8Z+80)$

The number of balls faced by B = 160

The strike-rate of B = $[\{(8Z+80)/160\} \times 100] = (5Z+50)$

So,

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$$(10X+25)/(5Z+50) = 5/6$$

$$60X + 150 = 25Z + 250$$

$$60X - 25Z = 100$$

$$12X - 5Z = 20 \dots\dots\dots(I)$$

Now,

The number of runs scored by C = X% of 800 = 8X

The number of balls faced by C = 100

The strike-rate of C = $\{(8X/100) \times 100\} = 8X$

The number of runs scored by D = Z% of 800 = 8Z

The number of balls faced by D = 320

The strike-rate of D = $\{(8Z/320) \times 100\} = 2.50Z$

So,

$$8X - 2.50Z = 30$$

$$16X - 5Z = 60 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$48X - 20Z = 80$$

$$48X - 15Z = 180$$

$$5Z = 100$$

$$Z = 20$$

Now, put the value of Z in equation (I), we get,

$$12X - 5Z = 20$$

$$12X - (5 \times 20) = 20$$

$$12X = 20 + 100$$

$$12X = 120$$

$$X = 10$$

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So, the percentage distribution of the total number of runs scored by each player,

$$A = (2X+5)\% = (2 \times 10+5)\% = (20+5)\% = 25\%$$

$$B = (Z+10)\% = (20+10)\% = 30\%$$

$$C = X\% = 10\%$$

$$D = Z\% = 20\%$$

$$E = \{100 - (25+30+10+20)\}\% = (100-85)\% = 15\%$$

So, $Y = 15$

For A

The number of runs scored = 25% of 800 = 200

The number of balls faced = 160

The strike-rate = $\{(200/160) \times 100\} \% = 125\%$

For B

The number of runs scored = 30% of 800 = 240

The number of balls faced = 160

The strike-rate = $\{(240/160) \times 100\} \% = 150\%$

For C

The number of runs scored = 10% of 800 = 80

The number of balls faced = 100

The strike-rate = $\{(80/100) \times 100\} \% = 80\%$

For D

The number of runs scored = 20% of 800 = 160

The number of balls faced = 320

The strike-rate = $\{(160/320) \times 100\} \% = 50\%$

For E

The number of runs scored = 15% of 800 = 120

The number of balls faced = 100

The strike-rate = $\{(120/100) \times 100\} \% = 120\%$

Name of the players	The number of runs scored	The number of balls faced	The strike-rate
A	200	160	125%
B	240	160	150%
C	80	100	80%
D	160	320	50%
E	120	100	120%
Total	800	840	95.23 (approx.)

Answer: B

Now, according to the question,

For A

The total number of runs scored = 200

The total number of runs scored through boundaries = $\frac{1}{4}$ of 200 = 50

Let the number of fours and sixes be scored by $8k$ and $3k$ respectively.

So, the total number of runs scored in fours = $(8k \times 4) = 32k$

And, the total number of runs scored in sixes = $(3k \times 6) = 18k$

Now,

$$32k + 18k = 50$$

$$50k = 50$$

$$k = 1$$

So, the number of balls faced for scoring boundary runs only = $(8k + 3k) = 11k = (11 \times 1) = 11$

Therefore, the required strike-rate = $\{(50/11) \times 100\} = 454.54$.

Hence, the correct answer is option B.

2. Questions

The number of runs scored by A = $(2X+5)\%$ of 800 = $(16X+40)$

The number of balls faced by A = 160

The strike-rate of A = $[\{(16X+40)/160\} \times 100] = (10X+25)$

The number of runs scored by B = $(Z+10)\%$ of 800 = $(8Z+80)$

The number of balls faced by B = 160

The strike-rate of B = $[\{(8Z+80)/160\} \times 100] = (5Z+50)$

So,

$$(10X+25)/(5Z+50) = 5/6$$

$$60X + 150 = 25Z + 250$$

$$60X - 25Z = 100$$

$$12X - 5Z = 20 \dots\dots\dots(I)$$

Now,

The number of runs scored by C = $X\%$ of 800 = $8X$

The number of balls faced by C = 100

The strike-rate of C = $\{(8X/100) \times 100\} = 8X$

The number of runs scored by D = $Z\%$ of 800 = $8Z$

The number of balls faced by D = 320

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The strike-rate of D = $\{(8Z/320) \times 100\} = 2.50Z$

So,

$$8X - 2.50Z = 30$$

$$16X - 5Z = 60 \dots\dots\dots (II)$$

By solving equations (I) and (II), we get,

$$48X - 20Z = 80$$

$$48X - 15Z = 180$$

$$5Z = 100$$

$$Z = 20$$

Now, put the value of Z in equation (I), we get,

$$12X - 5Z = 20$$

$$12X - (5 \times 20) = 20$$

$$12X = 20 + 100$$

$$12X = 120$$

$$X = 10$$

So, the percentage distribution of the total number of runs scored by each player,

$$A = (2X+5)\% = (2 \times 10 + 5)\% = (20+5)\% = 25\%$$

$$B = (Z+10)\% = (20+10)\% = 30\%$$

$$C = X\% = 10\%$$

$$D = Z\% = 20\%$$

$$E = \{100 - (25+30+10+20)\}\% = (100-85)\% = 15\%$$

$$\text{So, } Y = 15$$

For A

$$\text{The number of runs scored} = 25\% \text{ of } 800 = 200$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(200/160) \times 100\}\% = 125\%$$

For B

$$\text{The number of runs scored} = 30\% \text{ of } 800 = 240$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(240/160) \times 100\}\% = 150\%$$

For C

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The number of runs scored = 10% of 800 = 80

The number of balls faced = 100

The strike-rate = $\{(80/100) \times 100\} \% = 80\%$

For D

The number of runs scored = 20% of 800 = 160

The number of balls faced = 320

The strike-rate = $\{(160/320) \times 100\} \% = 50\%$

For E

The number of runs scored = 15% of 800 = 120

The number of balls faced = 100

The strike-rate = $\{(120/100) \times 100\} \% = 120\%$

Name of the players	The number of runs scored	The number of balls faced	The strike-rate
A	200	160	125%
B	240	160	150%
C	80	100	80%
D	160	320	50%
E	120	100	120%
Total	800	840	95.23 (approx.)

Answer: C

For C

The number of total balls faced (excluding dot balls) = (100-N)

The number of runs scored = 80

So, the strike rate (excluding dot balls) = 125% of 80 = 100

Now,

$$[\{80/(100-N)\} \times 100] = 100$$

$$80/(100-N) = 100/100$$

$$10000 - 100N = 8000$$

$$100N = 2000$$

$$N = 20$$

I: 20 is exactly divisible by only 5, but not 15.

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So, I is definitely not true.

II: 20 is not an odd integer.

So, II is definitely not true.

III: 20 is an even composite number.

So, III is definitely true.

Therefore, only III is definitely true.

Hence, the correct answer is option C.

3. Questions

The number of runs scored by A = $(2X+5)\%$ of 800 = $(16X+40)$

The number of balls faced by A = 160

The strike-rate of A = $[\{(16X+40)/160\} \times 100] = (10X+25)$

The number of runs scored by B = $(Z+10)\%$ of 800 = $(8Z+80)$

The number of balls faced by B = 160

The strike-rate of B = $[\{(8Z+80)/160\} \times 100] = (5Z+50)$

So,

$$(10X+25)/(5Z+50) = 5/6$$

$$60X + 150 = 25Z + 250$$

$$60X - 25Z = 100$$

$$12X - 5Z = 20 \dots\dots\dots(I)$$

Now,

The number of runs scored by C = $X\%$ of 800 = $8X$

The number of balls faced by C = 100

The strike-rate of C = $\{(8X/100) \times 100\} = 8X$

The number of runs scored by D = $Z\%$ of 800 = $8Z$

The number of balls faced by D = 320

The strike-rate of D = $\{(8Z/320) \times 100\} = 2.50Z$

So,

$$8X - 2.50Z = 30$$

$$16X - 5Z = 60 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$48X - 20Z = 80$$

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$$48X - 15Z = 180$$

$$5Z = 100$$

$$Z = 20$$

Now, put the value of Z in equation (I), we get,

$$12X - 5Z = 20$$

$$12X - (5 \times 20) = 20$$

$$12X = 20 + 100$$

$$12X = 120$$

$$X = 10$$

So, the percentage distribution of the total number of runs scored by each player,

$$A = (2X+5)\% = (2 \times 10 + 5)\% = (20+5)\% = 25\%$$

$$B = (Z+10)\% = (20+10)\% = 30\%$$

$$C = X\% = 10\%$$

$$D = Z\% = 20\%$$

$$E = \{100 - (25 + 30 + 10 + 20)\}\% = (100 - 85)\% = 15\%$$

$$\text{So, } Y = 15$$

For A

$$\text{The number of runs scored} = 25\% \text{ of } 800 = 200$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(200/160) \times 100\}\% = 125\%$$

For B

$$\text{The number of runs scored} = 30\% \text{ of } 800 = 240$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(240/160) \times 100\}\% = 150\%$$

For C

$$\text{The number of runs scored} = 10\% \text{ of } 800 = 80$$

$$\text{The number of balls faced} = 100$$

$$\text{The strike-rate} = \{(80/100) \times 100\}\% = 80\%$$

For D

$$\text{The number of runs scored} = 20\% \text{ of } 800 = 160$$

$$\text{The number of balls faced} = 320$$

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The strike-rate = $\{(160/320) \times 100\} \% = 50\%$

For E

The number of runs scored = 15% of 800 = 120

The number of balls faced = 100

The strike-rate = $\{(120/100) \times 100\} \% = 120\%$

Name of the players	The number of runs scored	The number of balls faced	The strike-rate
A	200	160	125%
B	240	160	150%
C	80	100	80%
D	160	320	50%
E	120	100	120%
Total	800	840	95.23 (approx.)

Answer: A

For F

Let the number of runs scored be m.

So,

$$\{(m/150) \times 100\} = (m-100)$$

$$m/150 = (m-100)/100$$

$$100m = 150m - 15000$$

$$50m = 15000$$

$$m = 300$$

So, the number of runs scored by F = m = 300

And, the number of runs scored by E = 120

Therefore, the required ratio = 120:300 = 2:5.

Hence, the correct answer is option A.

4. Questions

The number of runs scored by A = $(2X+5)\%$ of 800 = $(16X+40)$

The number of balls faced by A = 160

The strike-rate of A = $[\{(16X+40)/160\} \times 100] = (10X+25)$

The number of runs scored by B = $(Z+10)\%$ of 800 = $(8Z+80)$

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The number of balls faced by B = 160

The strike-rate of B = $\left[\frac{(8Z+80)}{160} \times 100\right] = (5Z+50)$

So,

$$(10X+25)/(5Z+50) = 5/6$$

$$60X + 150 = 25Z + 250$$

$$60X - 25Z = 100$$

$$12X - 5Z = 20 \dots\dots\dots(I)$$

Now,

The number of runs scored by C = X% of 800 = 8X

The number of balls faced by C = 100

The strike-rate of C = $\{(8X/100) \times 100\} = 8X$

The number of runs scored by D = Z% of 800 = 8Z

The number of balls faced by D = 320

The strike-rate of D = $\{(8Z/320) \times 100\} = 2.50Z$

So,

$$8X - 2.50Z = 30$$

$$16X - 5Z = 60 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$48X - 20Z = 80$$

$$48X - 15Z = 180$$

$$5Z = 100$$

$$Z = 20$$

Now, put the value of Z in equation (I), we get,

$$12X - 5Z = 20$$

$$12X - (5 \times 20) = 20$$

$$12X = 20 + 100$$

$$12X = 120$$

$$X = 10$$

So, the percentage distribution of the total number of runs scored by each player,

$$A = (2X+5)\% = (2 \times 10+5)\% = (20+5)\% = 25\%$$

$$B = (Z+10)\% = (20+10)\% = 30\%$$

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$$C = X\% = 10\%$$

$$D = Z\% = 20\%$$

$$E = \{100 - (25 + 30 + 10 + 20)\}\% = (100 - 85)\% = 15\%$$

$$\text{So, } Y = 15$$

For A

$$\text{The number of runs scored} = 25\% \text{ of } 800 = 200$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(200/160) \times 100\}\% = 125\%$$

For B

$$\text{The number of runs scored} = 30\% \text{ of } 800 = 240$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(240/160) \times 100\}\% = 150\%$$

For C

$$\text{The number of runs scored} = 10\% \text{ of } 800 = 80$$

$$\text{The number of balls faced} = 100$$

$$\text{The strike-rate} = \{(80/100) \times 100\}\% = 80\%$$

For D

$$\text{The number of runs scored} = 20\% \text{ of } 800 = 160$$

$$\text{The number of balls faced} = 320$$

$$\text{The strike-rate} = \{(160/320) \times 100\}\% = 50\%$$

For E

$$\text{The number of runs scored} = 15\% \text{ of } 800 = 120$$

$$\text{The number of balls faced} = 100$$

$$\text{The strike-rate} = \{(120/100) \times 100\}\% = 120\%$$

@QuizByMickey

Name of the players	The number of runs scored	The number of balls faced	The strike-rate
A	200	160	125%
B	240	160	150%
C	80	100	80%
D	160	320	50%
E	120	100	120%
Total	800	840	95.23 (approx.)

Answer: B

For D

The increased number of runs scored = $\{160 + (P\% \text{ of } 160)\} = (160 + 1.60P)$

The reduced number of balls faced = $(320 - Q)$

Now,

$$\{(160 + 1.60P) / (320 - Q)\} \times 100 = 100$$

$$(160 + 1.60P) / (320 - Q) = 100 / 100$$

$$(160 + 1.60P) / (320 - Q) = 1$$

$$160 + 1.60P = 320 - Q$$

$$Q + 1.60P = 160 \dots\dots\dots(i)$$

Also,

$$Q - P = 30 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$2.60P = 130$$

$$P = 50$$

Now, put the value of P in equation (i), we get,

$$Q + 1.60P = 160$$

$$Q + (1.60 \times 50) = 160$$

$$Q = 160 - 80$$

$$Q = 80$$

Therefore, $P < Q$.

Hence, the correct answer is option B.

5. Questions

@QuizByMickey

The number of runs scored by A = $(2X+5)\%$ of 800 = $(16X+40)$

The number of balls faced by A = 160

The strike-rate of A = $[(16X+40)/160] \times 100 = (10X+25)$

The number of runs scored by B = $(Z+10)\%$ of 800 = $(8Z+80)$

The number of balls faced by B = 160

The strike-rate of B = $[(8Z+80)/160] \times 100 = (5Z+50)$

So,

$$(10X+25)/(5Z+50) = 5/6$$

$$60X + 150 = 25Z + 250$$

$$60X - 25Z = 100$$

$$12X - 5Z = 20 \dots\dots\dots(I)$$

Now,

The number of runs scored by C = $X\%$ of 800 = $8X$

The number of balls faced by C = 100

The strike-rate of C = $\{(8X/100) \times 100\} = 8X$

The number of runs scored by D = $Z\%$ of 800 = $8Z$

The number of balls faced by D = 320

The strike-rate of D = $\{(8Z/320) \times 100\} = 2.50Z$

So,

$$8X - 2.50Z = 30$$

$$16X - 5Z = 60 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$48X - 20Z = 80$$

$$48X - 15Z = 180$$

$$5Z = 100$$

$$Z = 20$$

Now, put the value of Z in equation (I), we get,

$$12X - 5Z = 20$$

$$12X - (5 \times 20) = 20$$

$$12X = 20 + 100$$

$$12X = 120$$

@QuizByMickey

$$X = 10$$

So, the percentage distribution of the total number of runs scored by each player,

$$A = (2X+5)\% = (2 \times 10 + 5)\% = (20+5)\% = 25\%$$

$$B = (Z+10)\% = (20+10)\% = 30\%$$

$$C = X\% = 10\%$$

$$D = Z\% = 20\%$$

$$E = \{100 - (25 + 30 + 10 + 20)\}\% = (100 - 85)\% = 15\%$$

$$\text{So, } Y = 15$$

For A

$$\text{The number of runs scored} = 25\% \text{ of } 800 = 200$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(200/160) \times 100\}\% = 125\%$$

For B

$$\text{The number of runs scored} = 30\% \text{ of } 800 = 240$$

$$\text{The number of balls faced} = 160$$

$$\text{The strike-rate} = \{(240/160) \times 100\}\% = 150\%$$

For C

$$\text{The number of runs scored} = 10\% \text{ of } 800 = 80$$

$$\text{The number of balls faced} = 100$$

$$\text{The strike-rate} = \{(80/100) \times 100\}\% = 80\%$$

For D

$$\text{The number of runs scored} = 20\% \text{ of } 800 = 160$$

$$\text{The number of balls faced} = 320$$

$$\text{The strike-rate} = \{(160/320) \times 100\}\% = 50\%$$

For E

$$\text{The number of runs scored} = 15\% \text{ of } 800 = 120$$

$$\text{The number of balls faced} = 100$$

$$\text{The strike-rate} = \{(120/100) \times 100\}\% = 120\%$$

@QuizByMickey

Name of the players	The number of runs scored	The number of balls faced	The strike-rate
A	200	160	125%
B	240	160	150%
C	80	100	80%
D	160	320	50%
E	120	100	120%
Total	800	840	95.23 (approx.)

Answer: D

I: $(X+Z) = (10+20) = 30$

$2Y = (2*15) = 30$

So, I is correct.

II: $(Z-Y) = (20-15) = 5$

$(X/2) = (10/2) = 5$

So, II is correct.

III: $\{(X*Y)/10\} = \{(10*15)/10\} = (150/10) = 15$

$(Z-5) = (20-5) = 15$

So, III is correct.

Therefore, all I, II, and III follow.

Hence, the correct answer is option D.

6. Questions

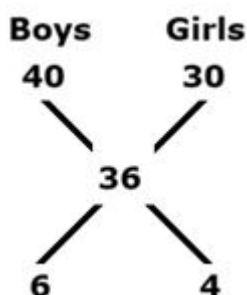
Class I

The average weight of all students (boys + girls) = $(7200/200) = 36$ kg

The average weight of each boy = 40 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



@QuizByMickey

So, the ratio of the number of boys and girls = $6:4 = 3:2$

The total number of students = 200

The number of boys = $\frac{3}{5}$ of 200 = 120

The number of girls = $\frac{2}{5}$ of 200 = 80

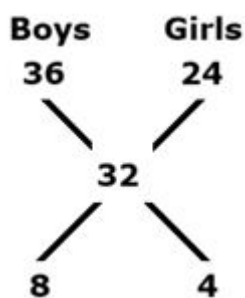
Class II

The average weight of all students (boys + girls) = $(3840/120) = 32$ kg

The average weight of each boy = 36 kg

The average weight of each girl = 24 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $8:4 = 2:1$

The number total number of students = 120

The number of boys = $\frac{2}{3}$ of 120 = 80

The number of girls = $\frac{1}{3}$ of 120 = 40

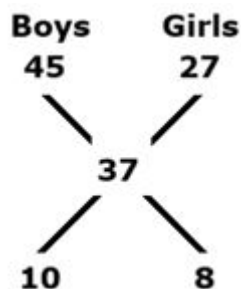
Class III

The average weight of all students (boys + girls) = $(13320/360) = 37$ kg

The average weight of each boy = 45 kg

The average weight of each girl = 27 kg

Now, using the allegation method, we get,



@QuizByMickey

So, the ratio of the number of boys and girls = $10:8 = 5:4$

The total number of students = 360

The number of boys = $\frac{5}{9}$ of 360 = 200

The number of girls = $\frac{4}{9}$ of $360 = 160$

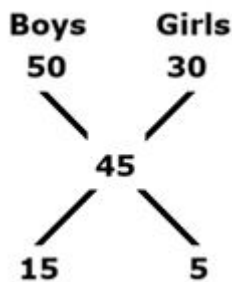
Class IV

The average weight of all students (boys + girls) = $(4500/100) = 45$ kg

The average weight of each boy = 50 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $15:5 = 3:1$

The total number of students = 100

The number of boys = $\frac{3}{4}$ of $100 = 75$

The number of girls = $\frac{1}{4}$ of $100 = 25$

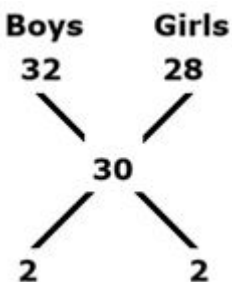
Class V

The average weight of all students (boys + girls) = $(12000/400) = 30$ kg

The average weight of each boy = 32 kg

The average weight of each girl = 28 kg

Now, using the allegation method, we get,



@QuizByMickey

So, the ratio of the number of boys and girls = $2:2 = 1:1$

The total number of students = 400

The number of boys = $\frac{1}{2}$ of $400 = 200$

The number of girls = $\frac{1}{2}$ of $400 = 200$

Class	The total number of students (boys + girls)	The number of boys	The number of girls
I	200	120	80
II	120	80	40
III	360	200	160
IV	100	75	25
V	400	200	200
Total	1180	675	505

Answer: D

Now, according to the question,

Class I

The total number of boys = 120

The total number of girls = 80

The ratio of the number of boys who belong to rural area and girls who belong to urban area = 1:1. Let it be k and k respectively.

So, the number of boys who belong to urban area = (120-k)

And, the number of girls who belong to rural area = (80-k)

Now,

$$(120-k)/(80-k) = 9/5$$

$$600 - 5k = 720 - 9k$$

$$4k = 120$$

$$k = 30$$

Therefore, the total number of girls who belong to rural area = (80-k) = (80-30) = 50.

Hence, the correct answer is option D.

7. Questions

@QuizByMickey

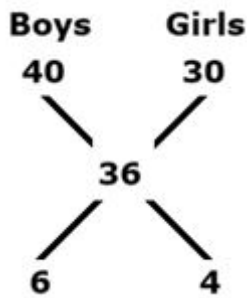
Class I

The average weight of all students (boys + girls) = (7200/200) = 36 kg

The average weight of each boy = 40 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = 6:4 = 3:2

The total number of students = 200

The number of boys = $\frac{3}{5}$ of 200 = 120

The number of girls = $\frac{2}{5}$ of 200 = 80

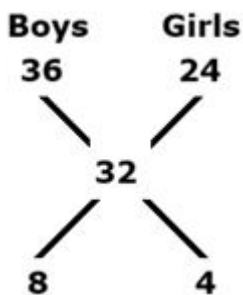
Class II

The average weight of all students (boys + girls) = $(3840/120) = 32$ kg

The average weight of each boy = 36 kg

The average weight of each girl = 24 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = 8:4 = 2:1

The number total number of students = 120

The number of boys = $\frac{2}{3}$ of 120 = 80

The number of girls = $\frac{1}{3}$ of 120 = 40

Class III

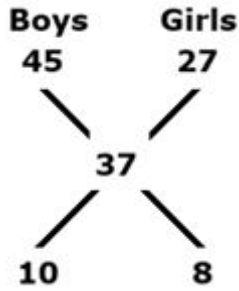
@QuizByMickey

The average weight of all students (boys + girls) = $(13320/360) = 37$ kg

The average weight of each boy = 45 kg

The average weight of each girl = 27 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $10:8 = 5:4$

The total number of students = 360

The number of boys = $\frac{5}{9}$ of 360 = 200

The number of girls = $\frac{4}{9}$ of 360 = 160

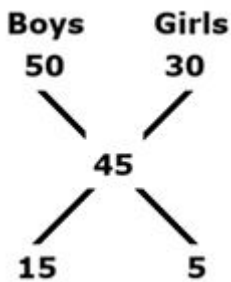
Class IV

The average weight of all students (boys + girls) = $(4500/100) = 45$ kg

The average weight of each boy = 50 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $15:5 = 3:1$

The total number of students = 100

The number of boys = $\frac{3}{4}$ of 100 = 75

The number of girls = $\frac{1}{4}$ of 100 = 25

Class V

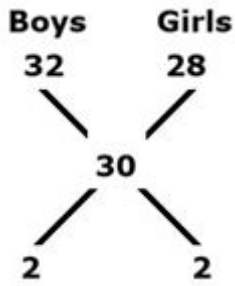
The average weight of all students (boys + girls) = $(12000/400) = 30$ kg

The average weight of each boy = 32 kg

The average weight of each girl = 28 kg

Now, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of boys and girls = $2:2 = 1:1$

The total number of students = 400

The number of boys = $\frac{1}{2}$ of 400 = 200

The number of girls = $\frac{1}{2}$ of 400 = 200

Class	The total number of students (boys + girls)	The number of boys	The number of girls
I	200	120	80
II	120	80	40
III	360	200	160
IV	100	75	25
V	400	200	200
Total	1180	675	505

Answer: A

Class II

The total number of students = 120

The total number of students who participated in sports = $\frac{5}{8}$ of 120 = 75 (**Note: $62.50\% = \frac{5}{8}$**)

The total number of students who participated in arts = $(120-75) = 45$

The total number of boys = 80

The number of boys who participated in arts = 30

The total number of girls = 40

The number of girls who participated in arts = $(45-30) = 15$

The number of girls who participated in sports = $(40-15) = 25$

Therefore, the required ratio = $25:15 = 5:3$.

Hence, the correct answer is option A.

8. Questions

Class I

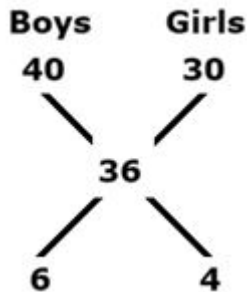
@QuizByMickey

The average weight of all students (boys + girls) = $(7200/200) = 36$ kg

The average weight of each boy = 40 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $6:4 = 3:2$

The total number of students = 200

The number of boys = $3/5$ of $200 = 120$

The number of girls = $2/5$ of $200 = 80$

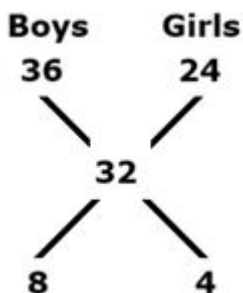
Class II

The average weight of all students (boys + girls) = $(3840/120) = 32$ kg

The average weight of each boy = 36 kg

The average weight of each girl = 24 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $8:4 = 2:1$

The number total number of students = 120

The number of boys = $2/3$ of $120 = 80$

The number of girls = $1/3$ of $120 = 40$

Class III

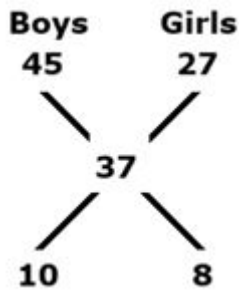
The average weight of all students (boys + girls) = $(13320/360) = 37$ kg

The average weight of each boy = 45 kg

The average weight of each girl = 27 kg

@QuizByMickey

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $10:8 = 5:4$

The total number of students = 360

The number of boys = $\frac{5}{9}$ of 360 = 200

The number of girls = $\frac{4}{9}$ of 360 = 160

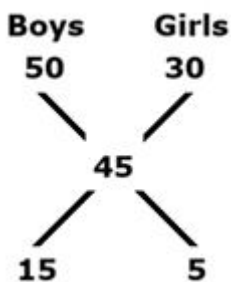
Class IV

The average weight of all students (boys + girls) = $(4500/100) = 45$ kg

The average weight of each boy = 50 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $15:5 = 3:1$

The total number of students = 100

The number of boys = $\frac{3}{4}$ of 100 = 75

The number of girls = $\frac{1}{4}$ of 100 = 25

Class V

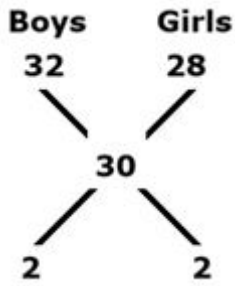
The average weight of all students (boys + girls) = $(12000/400) = 30$ kg

The average weight of each boy = 32 kg

The average weight of each girl = 28 kg

Now, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of boys and girls = $2:2 = 1:1$

The total number of students = 400

The number of boys = $\frac{1}{2}$ of 400 = 200

The number of girls = $\frac{1}{2}$ of 400 = 200

Class	The total number of students (boys + girls)	The number of boys	The number of girls
I	200	120	80
II	120	80	40
III	360	200	160
IV	100	75	25
V	400	200	200
Total	1180	675	505

Answer: E

Class III

The reduced number of boys = $(200-X)$

The increased number of girls = $(160+Y)$

The reduced total number of students = $(360-10) = 350$

The new number of boys = $\frac{1}{2}$ of 350 = 175

So,

$$(200-X) = 175$$

$$X = 25$$

The new number of girls = $\frac{1}{2}$ of 350 = 175

So,

$$(160+Y) = 175$$

$$Y = 15$$

Hence, the correct answer is option E.

@QuizByMickey

9. Questions

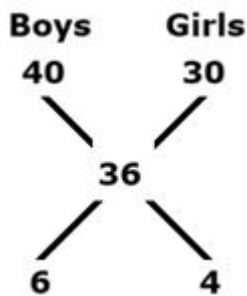
Class I

The average weight of all students (boys + girls) = $(7200/200) = 36$ kg

The average weight of each boy = 40 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = 6:4 = 3:2

The total number of students = 200

The number of boys = $3/5$ of 200 = 120

The number of girls = $2/5$ of 200 = 80

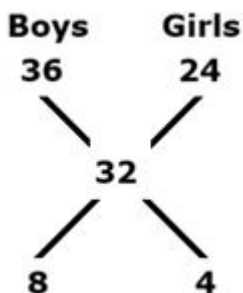
Class II

The average weight of all students (boys + girls) = $(3840/120) = 32$ kg

The average weight of each boy = 36 kg

The average weight of each girl = 24 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = 8:4 = 2:1

The number total number of students = 120

The number of boys = $2/3$ of 120 = 80

The number of girls = $1/3$ of 120 = 40

Class III

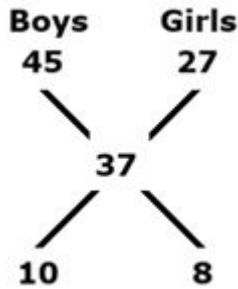
The average weight of all students (boys + girls) = $(13320/360) = 37$ kg

@QuizByMickey

The average weight of each boy = 45 kg

The average weight of each girl = 27 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = 10:8 = 5:4

The total number of students = 360

The number of boys = 5/9 of 360 = 200

The number of girls = 4/9 of 360 = 160

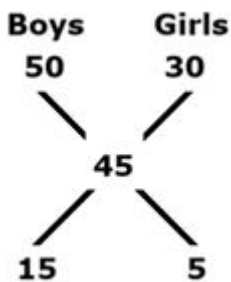
Class IV

The average weight of all students (boys + girls) = (4500/100) = 45 kg

The average weight of each boy = 50 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = 15:5 = 3:1

The total number of students = 100

The number of boys = 3/4 of 100 = 75

The number of girls = 1/4 of 100 = 25

Class V

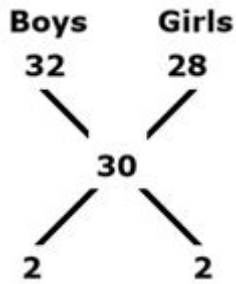
The average weight of all students (boys + girls) = (12000/400) = 30 kg

The average weight of each boy = 32 kg

The average weight of each girl = 28 kg

Now, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of boys and girls = 2:2 = 1:1

The total number of students = 400

The number of boys = $\frac{1}{2}$ of 400 = 200

The number of girls = $\frac{1}{2}$ of 400 = 200

Class	The total number of students (boys + girls)	The number of boys	The number of girls
I	200	120	80
II	120	80	40
III	360	200	160
IV	100	75	25
V	400	200	200
Total	1180	675	505

Answer: A

Class VI

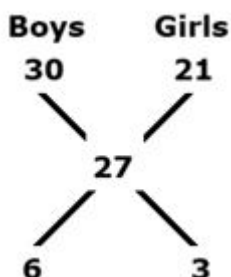
The total number of students = $\frac{3}{4}$ of 400 = 300

The average weight of all students (boys + girls) = $(8100/300) = 27$ kg

The average weight of each boy = 30 kg

The average weight of each girl = 21 kg

Now, using the allegation method,



So, the ratio of the number of boys and girls in class VI = 6:3 = 2:1.

Therefore, the difference between the number of boys and girls in class VI = $\frac{1}{3}$ of 300 = 100.

@QuizByMickey

Hence, the correct answer is option A.

10. Questions

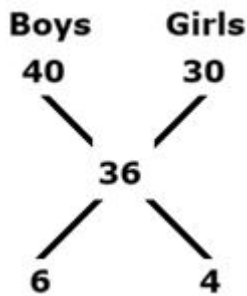
Class I

The average weight of all students (boys + girls) = $(7200/200) = 36$ kg

The average weight of each boy = 40 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $6:4 = 3:2$

The total number of students = 200

The number of boys = $3/5$ of 200 = 120

The number of girls = $2/5$ of 200 = 80

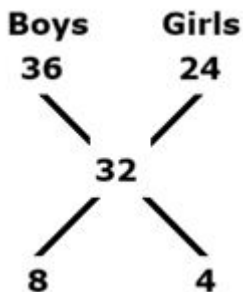
Class II

The average weight of all students (boys + girls) = $(3840/120) = 32$ kg

The average weight of each boy = 36 kg

The average weight of each girl = 24 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $8:4 = 2:1$

The number total number of students = 120

The number of boys = $2/3$ of 120 = 80

The number of girls = $1/3$ of 120 = 40

Class III

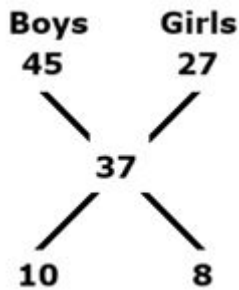
@QuizByMickey

The average weight of all students (boys + girls) = $(13320/360) = 37$ kg

The average weight of each boy = 45 kg

The average weight of each girl = 27 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $10:8 = 5:4$

The total number of students = 360

The number of boys = $5/9$ of 360 = 200

The number of girls = $4/9$ of 360 = 160

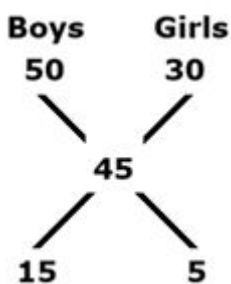
Class IV

The average weight of all students (boys + girls) = $(4500/100) = 45$ kg

The average weight of each boy = 50 kg

The average weight of each girl = 30 kg

Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = $15:5 = 3:1$

The total number of students = 100

The number of boys = $3/4$ of 100 = 75

The number of girls = $1/4$ of 100 = 25

Class V

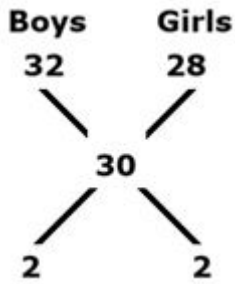
The average weight of all students (boys + girls) = $(12000/400) = 30$ kg

The average weight of each boy = 32 kg

The average weight of each girl = 28 kg

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Now, using the allegation method, we get,



So, the ratio of the number of boys and girls = 2:2 = 1:1

The total number of students = 400

The number of boys = $\frac{1}{2}$ of 400 = 200

The number of girls = $\frac{1}{2}$ of 400 = 200

Class	The total number of students (boys + girls)	The number of boys	The number of girls
I	200	120	80
II	120	80	40
III	360	200	160
IV	100	75	25
V	400	200	200
Total	1180	675	505

Answer: B

Class IV

The ratio of the average age of each boy and each girl = 125:100 = 5:4. Let it be 5n years and 4n years.

The total number of students (boys + girls) = 100

So, the total age of students (boys + girls) = $(9.50 \times 100) = 950$ years

The total number of boys = 75

So, the total age of all boys = $(5n \times 75) = 375n$ years

The total number of girls = 25

So, the total age of all girls = $(4n \times 25) = 100n$ years

Now,

$$375n + 100n = 950$$

$$475n = 950$$

$$n = 2$$

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Therefore, the total age of all boys = $375n = (375 \times 2) = 750$ years.

Hence, the correct answer is option B.

11. Questions

Shop P and Q

Let the total number of cakes sold by all five shops together be 100k.

So, the total number of cakes sold by shop P = 24% of 100k = 24k

The number of chocolate cakes sold by shop P = 120

The sum of the number of vanilla and chocolate cakes sold by shop P = $(24k - 120)$

The number of vanilla cakes sold by shop P = $\frac{2}{3}$ of $(24k - 120) = (16k - 80)$

The number of strawberry cakes sold by shop P = $\frac{1}{3}$ of $(24k - 120) = (8k - 40)$

Now,

The total number of cakes sold by shop Q = 16% of 100k = 16k

The number of chocolate cakes sold by shop Q = 64

The sum of the number of vanilla and chocolate cakes sold by shop Q = $(16k - 64)$

The number of vanilla cakes sold by shop Q = $\frac{3}{4}$ of $(16k - 64) = (12k - 48)$

The number of strawberry cakes sold by shop Q = $\frac{1}{4}$ of $(16k - 64) = (4k - 16)$

So,

$$(16k - 80) - (4k - 16) = 56$$

$$16k - 136 = 4k - 16$$

$$12k = 120$$

$$k = 10$$

Therefore, the total number of cakes sold by all five shops together = $100k = (100 \times 10) = 1000$.

Shop P

The total number of cakes sold = 24% of 1000 = 240

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(240 - 120) = 120$

The number of vanilla cakes sold = $\frac{2}{3}$ of 120 = 80

The number of strawberry cakes sold = $\frac{1}{3}$ of 120 = 40

Shop Q

The total number of cakes sold = 16% of 1000 = 160

The number of chocolate cakes sold = 64

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The sum of the number of vanilla and strawberry cakes sold = $(160-64) = 96$

The number of vanilla cakes sold = $\frac{3}{4}$ of $96 = 72$

The number of strawberry cakes sold = $\frac{1}{4}$ of $96 = 24$

Shop R

The total number of cakes sold = 32% of $1000 = 320$

The number of chocolate cakes sold = 240

The sum of the number of vanilla and strawberry cakes sold = $(320-240) = 80$

The number of vanilla cakes sold = $\frac{3}{5}$ of $80 = 48$

The number of strawberry cakes sold = $\frac{2}{5}$ of $80 = 32$

Shop S

The total number of cakes sold = 20% of $1000 = 200$

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(200-120) = 80$

The number of vanilla cakes sold = $\frac{1}{2}$ of $80 = 40$

The number of strawberry cakes sold = $\frac{1}{2}$ of $80 = 40$

Shop T

The total number of cakes sold = 8% of $1000 = 80$

The number of chocolate cakes sold = 56

The sum of the number of vanilla and strawberry cakes sold = $(80-56) = 24$

The number of vanilla cakes sold = $\frac{5}{8}$ of $24 = 15$

The number of strawberry cakes sold = $\frac{3}{8}$ of $24 = 9$

Shop	The total number of cakes sold	The number of chocolate cakes sold	The number of vanilla cakes sold	The number of strawberry cakes sold
P	240	120	80	40
Q	160	64	72	24
R	320	240	48	32
S	200	120	40	40
T	80	56	15	9
Total	1000	600	255	145

Answer: B

Now, according to the question,

The ratio of the total number of cakes unsold by shop R and S = $100:75 = 4:3$. Let it be $4a$ and $3a$ respectively.

So, the total number of cakes made (sold + unsold) by,

$$\text{Shop R} = (320+4a)$$

$$\text{Shop S} = (200+3a)$$

Now,

$$(320+4a)/(200+3a) = 20/13$$

$$4160 + 52a = 4000 + 60a$$

$$8a = 160$$

$$a = 20$$

I = The total number of cakes sold by shop R = 320

The total number of cakes unsold by shop R = $4a = (4 \times 20) = 80$

The total number of cakes made (sold + unsold) by shop R = $(320+80) = 400$

The required percentage = $\{(320/400) \times 100\} \% = 80\%$

So, M = 80.

II = The total number of cakes sold by shop S = 200

The total number of cakes unsold by shop S = $3a = (3 \times 20) = 60$

The required difference = $(200-60) = 140$

So, N = 140.

Therefore, $I < II$.

Hence, the correct answer is option B.

12. Questions

Shop P and Q

Let the total number of cakes sold by all five shops together be 100k.

So, the total number of cakes sold by shop P = 24% of 100k = 24k

The number of chocolate cakes sold by shop P = 120

The sum of the number of vanilla and chocolate cakes sold by shop P = $(24k-120)$

The number of vanilla cakes sold by shop P = $2/3$ of $(24k-120) = (16k-80)$

The number of strawberry cakes sold by shop P = $1/3$ of $(24k-120) = (8k-40)$

Now,

The total number of cakes sold by shop Q = 16% of 100k = 16k

The number of chocolate cakes sold by shop Q = 64

The sum of the number of vanilla and chocolate cakes sold by shop Q = (16k-64)

The number of vanilla cakes sold by shop Q = $\frac{3}{4}$ of (16k-64) = (12k-48)

The number of strawberry cakes sold by shop Q = $\frac{1}{4}$ of (16k-64) = (4k-16)

So,

$$(16k-80) - (4k-16) = 56$$

$$16k - 136 = 4k - 16$$

$$12k = 120$$

$$k = 10$$

Therefore, the total number of cakes sold by all five shops together = 100k = (100×10) = 1000.

Shop P

The total number of cakes sold = 24% of 1000 = 240

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = (240-120) = 120

The number of vanilla cakes sold = $\frac{2}{3}$ of 120 = 80

The number of strawberry cakes sold = $\frac{1}{3}$ of 120 = 40

Shop Q

The total number of cakes sold = 16% of 1000 = 160

The number of chocolate cakes sold = 64

The sum of the number of vanilla and strawberry cakes sold = (160-64) = 96

The number of vanilla cakes sold = $\frac{3}{4}$ of 96 = 72

The number of strawberry cakes sold = $\frac{1}{4}$ of 96 = 24

Shop R

The total number of cakes sold = 32% of 1000 = 320

The number of chocolate cakes sold = 240

The sum of the number of vanilla and strawberry cakes sold = (320-240) = 80

The number of vanilla cakes sold = $\frac{3}{5}$ of 80 = 48

The number of strawberry cakes sold = $\frac{2}{5}$ of 80 = 32

Shop S

The total number of cakes sold = 20% of 1000 = 200

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The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(200-120) = 80$

The number of vanilla cakes sold = $1/1$ of $80 = 40$

The number of strawberry cakes sold = $1/1$ of $80 = 40$

Shop T

The total number of cakes sold = 8% of $1000 = 80$

The number of chocolate cakes sold = 56

The sum of the number of vanilla and strawberry cakes sold = $(80-56) = 24$

The number of vanilla cakes sold = $5/8$ of $24 = 15$

The number of strawberry cakes sold = $3/8$ of $24 = 9$

Shop	The total number of cakes sold	The number of chocolate cakes sold	The number of vanilla cakes sold	The number of strawberry cakes sold
P	240	120	80	40
Q	160	64	72	24
R	320	240	48	32
S	200	120	40	40
T	80	56	15	9
Total	1000	600	255	145

Answer: A

Shop P

The total number of cakes sold = 240

The total number of cakes sold to males = $3/4$ of $240 = 180$

The total number of cakes sold to females = $1/4$ of $240 = 60$

The number of vanilla cakes sold to females = $2/6$ of $60 = 1/3$ of $60 = 20$

The total number of vanilla cakes sold (males + females) = 80

So, the number of vanilla cakes sold to males = $(80-20) = 60$

Therefore, the required percentage = $\{(60/80) \times 100\} \% = 75\%$.

Hence, the correct answer is option A.

13. Questions

Shop P and Q

Let the total number of cakes sold by all five shops together be $100k$.

So, the total number of cakes sold by shop P = 24% of $100k = 24k$

The number of chocolate cakes sold by shop P = 120

The sum of the number of vanilla and chocolate cakes sold by shop P = $(24k - 120)$

The number of vanilla cakes sold by shop P = $\frac{2}{3}$ of $(24k - 120) = (16k - 80)$

The number of strawberry cakes sold by shop P = $\frac{1}{3}$ of $(24k - 120) = (8k - 40)$

Now,

The total number of cakes sold by shop Q = 16% of $100k = 16k$

The number of chocolate cakes sold by shop Q = 64

The sum of the number of vanilla and chocolate cakes sold by shop Q = $(16k - 64)$

The number of vanilla cakes sold by shop Q = $\frac{3}{4}$ of $(16k - 64) = (12k - 48)$

The number of strawberry cakes sold by shop Q = $\frac{1}{4}$ of $(16k - 64) = (4k - 16)$

So,

$$(16k - 80) - (4k - 16) = 56$$

$$16k - 136 = 4k - 16$$

$$12k = 120$$

$$k = 10$$

Therefore, the total number of cakes sold by all five shops together = $100k = (100 \times 10) = 1000$.

Shop P

The total number of cakes sold = 24% of $1000 = 240$

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(240 - 120) = 120$

The number of vanilla cakes sold = $\frac{2}{3}$ of $120 = 80$

The number of strawberry cakes sold = $\frac{1}{3}$ of $120 = 40$

Shop Q

The total number of cakes sold = 16% of $1000 = 160$

The number of chocolate cakes sold = 64

The sum of the number of vanilla and strawberry cakes sold = $(160 - 64) = 96$

The number of vanilla cakes sold = $\frac{3}{4}$ of $96 = 72$

The number of strawberry cakes sold = $\frac{1}{4}$ of $96 = 24$

Shop R

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The total number of cakes sold = 32% of 1000 = 320

The number of chocolate cakes sold = 240

The sum of the number of vanilla and strawberry cakes sold = (320-240) = 80

The number of vanilla cakes sold = $\frac{3}{5}$ of 80 = 48

The number of strawberry cakes sold = $\frac{2}{5}$ of 80 = 32

Shop S

The total number of cakes sold = 20% of 1000 = 200

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = (200-120) = 80

The number of vanilla cakes sold = $\frac{1}{2}$ of 80 = 40

The number of strawberry cakes sold = $\frac{1}{2}$ of 80 = 40

Shop T

The total number of cakes sold = 8% of 1000 = 80

The number of chocolate cakes sold = 56

The sum of the number of vanilla and strawberry cakes sold = (80-56) = 24

The number of vanilla cakes sold = $\frac{5}{8}$ of 24 = 15

The number of strawberry cakes sold = $\frac{3}{8}$ of 24 = 9

Shop	The total number of cakes sold	The number of chocolate cakes sold	The number of vanilla cakes sold	The number of strawberry cakes sold
P	240	120	80	40
Q	160	64	72	24
R	320	240	48	32
S	200	120	40	40
T	80	56	15	9
Total	1000	600	255	145

Answer: C

I: The number of vanilla cakes sold by,

Shop P = 80

Shop Q = 72

The required percentage = $\left[\frac{(80-72)}{80} \times 100\right]\% = \left[\frac{(8)}{80} \times 100\right]\% = 10\%$.

So, I is definitely true.

II: The number of chocolate cakes sold by shop S = 120

The number of strawberry cakes sold by shop S = 40

The required ratio = $120:40 = 3:1$.

So, II is definitely true.

III: The total number of cakes sold by,

Shop R = 320

Shop T = 80

The required difference = $(320-80) = 240$

So, III is definitely not true.

Therefore, both I and II are definitely true.

Hence, the correct answer is option C.

14. Questions

Shop P and Q

Let the total number of cakes sold by all five shops together be 100k.

So, the total number of cakes sold by shop P = 24% of $100k = 24k$

The number of chocolate cakes sold by shop P = 120

The sum of the number of vanilla and chocolate cakes sold by shop P = $(24k-120)$

The number of vanilla cakes sold by shop P = $\frac{2}{3}$ of $(24k-120) = (16k-80)$

The number of strawberry cakes sold by shop P = $\frac{1}{3}$ of $(24k-120) = (8k-40)$

Now,

The total number of cakes sold by shop Q = 16% of $100k = 16k$

The number of chocolate cakes sold by shop Q = 64

The sum of the number of vanilla and chocolate cakes sold by shop Q = $(16k-64)$

The number of vanilla cakes sold by shop Q = $\frac{3}{4}$ of $(16k-64) = (12k-48)$

The number of strawberry cakes sold by shop Q = $\frac{1}{4}$ of $(16k-64) = (4k-16)$

So,

$$(16k-80) - (4k-16) = 56$$

$$16k - 136 = 4k - 16$$

$$12k = 120$$

$$k = 10$$

Therefore, the total number of cakes sold by all five shops together = $100k = (100 \times 10) = 1000$.

Shop P

The total number of cakes sold = 24% of 1000 = 240

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(240 - 120) = 120$

The number of vanilla cakes sold = $\frac{2}{3}$ of 120 = 80

The number of strawberry cakes sold = $\frac{1}{3}$ of 120 = 40

Shop Q

The total number of cakes sold = 16% of 1000 = 160

The number of chocolate cakes sold = 64

The sum of the number of vanilla and strawberry cakes sold = $(160 - 64) = 96$

The number of vanilla cakes sold = $\frac{3}{4}$ of 96 = 72

The number of strawberry cakes sold = $\frac{1}{4}$ of 96 = 24

Shop R

The total number of cakes sold = 32% of 1000 = 320

The number of chocolate cakes sold = 240

The sum of the number of vanilla and strawberry cakes sold = $(320 - 240) = 80$

The number of vanilla cakes sold = $\frac{3}{5}$ of 80 = 48

The number of strawberry cakes sold = $\frac{2}{5}$ of 80 = 32

Shop S

The total number of cakes sold = 20% of 1000 = 200

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(200 - 120) = 80$

The number of vanilla cakes sold = $\frac{1}{1}$ of 80 = 40

The number of strawberry cakes sold = $\frac{1}{1}$ of 80 = 40

Shop T

The total number of cakes sold = 8% of 1000 = 80

The number of chocolate cakes sold = 56

The sum of the number of vanilla and strawberry cakes sold = $(80 - 56) = 24$

The number of vanilla cakes sold = $\frac{5}{8}$ of 24 = 15

The number of strawberry cakes sold = $\frac{3}{8}$ of 24 = 9

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Shop	The total number of cakes sold	The number of chocolate cakes sold	The number of vanilla cakes sold	The number of strawberry cakes sold
P	240	120	80	40
Q	160	64	72	24
R	320	240	48	32
S	200	120	40	40
T	80	56	15	9
Total	1000	600	255	145

Answer: E

Shop U

The total number of cakes sold = $(200-20) = 180$

The number of strawberry cakes sold = 20

The sum of the number of chocolate and vanilla cakes sold = $(180-20) = 160$

Therefore, the number of chocolate cakes sold = $\{(160+20)/2\} = (180/2) = 90$.

Hence, the correct answer is option E.

15. Questions

Shop P and Q

Let the total number of cakes sold by all five shops together be 100k.

So, the total number of cakes sold by shop P = 24% of 100k = 24k

The number of chocolate cakes sold by shop P = 120

The sum of the number of vanilla and chocolate cakes sold by shop P = $(24k-120)$

The number of vanilla cakes sold by shop P = $2/3$ of $(24k-120) = (16k-80)$

The number of strawberry cakes sold by shop P = $1/3$ of $(24k-120) = (8k-40)$

Now,

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The total number of cakes sold by shop Q = 16% of 100k = 16k

The number of chocolate cakes sold by shop Q = 64

The sum of the number of vanilla and chocolate cakes sold by shop Q = $(16k-64)$

The number of vanilla cakes sold by shop Q = $3/4$ of $(16k-64) = (12k-48)$

The number of strawberry cakes sold by shop Q = $1/4$ of $(16k-64) = (4k-16)$

So,

$$(16k-80) - (4k-16) = 56$$

$$16k - 136 = 4k - 16$$

$$12k = 120$$

$$k = 10$$

Therefore, the total number of cakes sold by all five shops together = $100k = (100 \times 10) = 1000$.

Shop P

The total number of cakes sold = 24% of 1000 = 240

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(240-120) = 120$

The number of vanilla cakes sold = $\frac{2}{3}$ of 120 = 80

The number of strawberry cakes sold = $\frac{1}{3}$ of 120 = 40

Shop Q

The total number of cakes sold = 16% of 1000 = 160

The number of chocolate cakes sold = 64

The sum of the number of vanilla and strawberry cakes sold = $(160-64) = 96$

The number of vanilla cakes sold = $\frac{3}{4}$ of 96 = 72

The number of strawberry cakes sold = $\frac{1}{4}$ of 96 = 24

Shop R

The total number of cakes sold = 32% of 1000 = 320

The number of chocolate cakes sold = 240

The sum of the number of vanilla and strawberry cakes sold = $(320-240) = 80$

The number of vanilla cakes sold = $\frac{3}{5}$ of 80 = 48

The number of strawberry cakes sold = $\frac{2}{5}$ of 80 = 32

Shop S

The total number of cakes sold = 20% of 1000 = 200

The number of chocolate cakes sold = 120

The sum of the number of vanilla and strawberry cakes sold = $(200-120) = 80$

The number of vanilla cakes sold = $\frac{1}{1}$ of 80 = 40

The number of strawberry cakes sold = $\frac{1}{1}$ of 80 = 40

Shop T

The total number of cakes sold = 8% of 1000 = 80

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The number of chocolate cakes sold = 56

The sum of the number of vanilla and strawberry cakes sold = $(80-56) = 24$

The number of vanilla cakes sold = $\frac{5}{8}$ of $24 = 15$

The number of strawberry cakes sold = $\frac{3}{8}$ of $24 = 9$

Shop	The total number of cakes sold	The number of chocolate cakes sold	The number of vanilla cakes sold	The number of strawberry cakes sold
P	240	120	80	40
Q	160	64	72	24
R	320	240	48	32
S	200	120	40	40
T	80	56	15	9
Total	1000	600	255	145

Answer: D

Let the price of each vanilla cake in shop Q and T be Rs. 3P and Rs. 5P respectively.

The total number of vanilla cakes sold by,

Shop Q = 72

Shop T = 15

So, the total revenue generated by,

Shop Q = $(72 \times 3P) = \text{Rs. } 216P$

Shop T = $(15 \times 5P) = \text{Rs. } 75P$

Now,

$216P - 75P = 1410$

$141P = 1410$

$P = 10$

Therefore, required difference = $(5P-3P) = 2P = (2 \times 10) = \text{Rs. } 20$.

Hence, the correct answer is option D.

16. Questions

Bank X

The total number of people who have account = 2000

The total number of people who have savings account = 60% of 2000 = 1200

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The total number of people who have current account = 40% of 2000 = 800

Let the number of males who have savings account and females who have current account be 3m and m respectively.

The ratio of the number of females who have savings account and males who have current account = 3:5.
 Let it be 3n and 5n respectively.

Now,

$$3m + 3n = 1200$$

$$m + n = 400 \dots\dots\dots(I)$$

And,

$$m + 5n = 800 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$4n = 400$$

$$n = 100$$

Now, put the value of n in equation (I), we get,

$$m + n = 400$$

$$m + 100 = 400$$

$$m = 300$$

Therefore,

$$\text{The number of males who have savings account} = 3m = (3 \times 300) = 900$$

$$\text{The number of females who have savings account} = 3n = (3 \times 100) = 300$$

$$\text{The number of males who have current account} = 5n = (5 \times 100) = 500$$

$$\text{The number of females who have current account} = m = 300$$

Bank Y

The total number of people who have savings account = 1000

The total number of people who have current account = $(1000/2) = 500$

The total number of people who have account = $(1000+500) = 1500$

The ratio of the number of males who have savings account and females who have current account = 6:1.
 Let it be 6a and a respectively. **(Note: 16.67% = 1/6)**

Also, the ratio of the number of females who have savings account and males who have current account = 1:1. Let it be b and b respectively.

Now,

$$6a + b = 1000 \dots\dots\dots(III)$$

And,

$$a + b = 500 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$6a + b = 1000$$

$$6a + 6b = 3000$$

$$5b = 2000$$

$$b = 400$$

Now, put the value of b in equation (III), we get,

$$6a + b = 1000$$

$$6a + 400 = 1000$$

$$6a = 600$$

$$a = 100$$

Therefore,

The number of males who have savings account = $6a = (6 \times 100) = 600$

The number of females who have savings account = $b = 400$

The number of males who have current account = $b = 400$

The number of females who have current account = $a = 100$

Savings Account

Bank	The total number of people who have savings account	The number of males who have savings account	The number of females who have savings account
X	1200	900	300
Y	1000	600	400
Total	2200	1500	700

Current Account

Bank	The total number of people who have current account	The number of males who have current account	The number of females who have current account
X	800	500	300
Y	500	400	100
Total	1300	900	400

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Answer: B

Now, according to the question,

Bank X

The total number of males who have accounts = 1400

The total number of males whose account is active = $\frac{5}{7}$ of 1400 = 1000

The total number of males whose account is dormant = $\frac{2}{7}$ of 1400 = 400

The total number of males who have savings account = 900

The number of males whose savings account is dormant = $\frac{1}{3}$ of 900 = 300

So, the number of males whose current account is dormant = $(400 - 300) = 100$

Also, the total number of males who have current account = 500

Therefore, the required percentage = $\left\{ \left(\frac{100}{500} \right) \times 100 \right\} \% = 20\%$.

Hence, the correct answer is option B.

17. Questions

Bank X

The total number of people who have account = 2000

The total number of people who have savings account = 60% of 2000 = 1200

The total number of people who have current account = 40% of 2000 = 800

Let the number of males who have savings account and females who have current account be $3m$ and m respectively.

The ratio of the number of females who have savings account and males who have current account = 3:5.
Let it be $3n$ and $5n$ respectively.

Now,

$$3m + 3n = 1200$$

$$m + n = 400 \dots \dots \dots (I)$$

And,

$$m + 5n = 800 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$4n = 400$$

$$n = 100$$

Now, put the value of n in equation (I), we get,

$$m + n = 400$$

$$m + 100 = 400$$

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$$m = 300$$

Therefore,

$$\text{The number of males who have savings account} = 3m = (3 \times 300) = 900$$

$$\text{The number of females who have savings account} = 3n = (3 \times 100) = 300$$

$$\text{The number of males who have current account} = 5n = (5 \times 100) = 500$$

$$\text{The number of females who have current account} = m = 300$$

Bank Y

$$\text{The total number of people who have savings account} = 1000$$

$$\text{The total number of people who have current account} = (1000/2) = 500$$

$$\text{The total number of people who have account} = (1000+500) = 1500$$

The ratio of the number of males who have savings account and females who have current account = 6:1.

Let it be $6a$ and a respectively. (Note: $16.67\% = 1/6$)

Also, the ratio of the number of females who have savings account and males who have current account = 1:1. Let it be b and b respectively.

Now,

$$6a + b = 1000 \dots \dots \dots \text{(III)}$$

And,

$$a + b = 500 \dots \dots \dots \text{(IV)}$$

By solving equations (III) and (IV), we get,

$$6a + b = 1000$$

$$6a + 6b = 3000$$

$$5b = 2000$$

$$b = 400$$

Now, put the value of b in equation (III), we get,

$$6a + b = 1000$$

$$6a + 400 = 1000$$

$$6a = 600$$

$$a = 100$$

Therefore,

$$\text{The number of males who have savings account} = 6a = (6 \times 100) = 600$$

$$\text{The number of females who have savings account} = b = 400$$

$$\text{The number of males who have current account} = b = 400$$

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The number of females who have current account = a = 100

Savings Account

Bank	The total number of people who have savings account	The number of males who have savings account	The number of females who have savings account
X	1200	900	300
Y	1000	600	400
Total	2200	1500	700

Current Account

Bank	The total number of people who have current account	The number of males who have current account	The number of females who have current account
X	800	500	300
Y	500	400	100
Total	1300	900	400

Answer: C

Bank Y

The total number of males who have savings account = 600

The number of male senior citizens who have savings account = 25% of 600 = 150

The number of male adults who have savings account = (600-150) = 450

The total number of females who have savings account = 400

The number of female senior citizens who have savings account = (150-70) = 80

The number of female adults who have savings account = (400-80) = 320

Therefore, the total number of adults who have savings account = (450+320) = 770.

Hence, the correct answer is option C.

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18. Questions

Bank X

The total number of people who have account = 2000

The total number of people who have savings account = 60% of 2000 = 1200

The total number of people who have current account = 40% of 2000 = 800

Let the number of males who have savings account and females who have current account be 3m and m respectively.

The ratio of the number of females who have savings account and males who have current account = 3:5.
Let it be $3n$ and $5n$ respectively.

Now,

$$3m + 3n = 1200$$

$$m + n = 400 \dots\dots\dots(I)$$

And,

$$m + 5n = 800 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$4n = 400$$

$$n = 100$$

Now, put the value of n in equation (I), we get,

$$m + n = 400$$

$$m + 100 = 400$$

$$m = 300$$

Therefore,

$$\text{The number of males who have savings account} = 3m = (3 \times 300) = 900$$

$$\text{The number of females who have savings account} = 3n = (3 \times 100) = 300$$

$$\text{The number of males who have current account} = 5n = (5 \times 100) = 500$$

$$\text{The number of females who have current account} = m = 300$$

Bank Y

$$\text{The total number of people who have savings account} = 1000$$

$$\text{The total number of people who have current account} = (1000/2) = 500$$

$$\text{The total number of people who have account} = (1000+500) = 1500$$

The ratio of the number of males who have savings account and females who have current account = 6:1.

Let it be $6a$ and a respectively. (Note: $16.67\% = 1/6$)

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Also, the ratio of the number of females who have savings account and males who have current account = 1:1. Let it be b and b respectively.

Now,

$$6a + b = 1000 \dots\dots\dots(III)$$

And,

$$a + b = 500 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$6a + b = 1000$$

$$6a + 6b = 3000$$

$$5b = 2000$$

$$b = 400$$

Now, put the value of b in equation (III), we get,

$$6a + b = 1000$$

$$6a + 400 = 1000$$

$$6a = 600$$

$$a = 100$$

Therefore,

The number of males who have savings account = $6a = (6 \times 100) = 600$

The number of females who have savings account = $b = 400$

The number of males who have current account = $b = 400$

The number of females who have current account = $a = 100$

Savings Account

Bank	The total number of people who have savings account	The number of males who have savings account	The number of females who have savings account
X	1200	900	300
Y	1000	600	400
Total	2200	1500	700

Current Account

Bank	The total number of people who have current account	The number of males who have current account	The number of females who have current account
X	800	500	300
Y	500	400	100
Total	1300	900	400

Answer: D

Using statement I and II together

The ratio of the total number of people who have savings and current account in bank Z = $75:25 = 3:1$

So, the total number of people who have account in bank Z = $\{(1200/2) \times 4\} = 2400$

Therefore, the required percentage = $\{(400/2400) \times 100\} \% = 16.67\%$.

Hence, both statement I and II together are necessary.

19. Questions

Bank X

The total number of people who have account = 2000

The total number of people who have savings account = 60% of 2000 = 1200

The total number of people who have current account = 40% of 2000 = 800

Let the number of males who have savings account and females who have current account be 3m and m respectively.

The ratio of the number of females who have savings account and males who have current account = 3:5.
Let it be 3n and 5n respectively.

Now,

$$3m + 3n = 1200$$

$$m + n = 400 \dots\dots\dots(I)$$

And,

$$m + 5n = 800 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$4n = 400$$

$$n = 100$$

Now, put the value of n in equation (I), we get,

$$m + n = 400$$

$$m + 100 = 400$$

$$m = 300$$

Therefore,

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The number of males who have savings account = 3m = $(3 \times 300) = 900$

The number of females who have savings account = 3n = $(3 \times 100) = 300$

The number of males who have current account = 5n = $(5 \times 100) = 500$

The number of females who have current account = m = 300

Bank Y

The total number of people who have savings account = 1000

The total number of people who have current account = $(1000/2) = 500$

The total number of people who have account = $(1000+500) = 1500$

The ratio of the number of males who have savings account and females who have current account = 6:1.
 Let it be $6a$ and a respectively. (**Note: $16.67\% = 1/6$**)

Also, the ratio of the number of females who have savings account and males who have current account = 1:1. Let it be b and b respectively.

Now,

$$6a + b = 1000 \dots\dots\dots(III)$$

And,

$$a + b = 500 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$6a + b = 1000$$

$$6a + 6b = 3000$$

$$5b = 2000$$

$$b = 400$$

Now, put the value of b in equation (III), we get,

$$6a + b = 1000$$

$$6a + 400 = 1000$$

$$6a = 600$$

$$a = 100$$

Therefore,

The number of males who have savings account = $6a = (6 \times 100) = 600$

The number of females who have savings account = $b = 400$

The number of males who have current account = $b = 400$

The number of females who have current account = $a = 100$

Savings Account

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Bank	The total number of people who have savings account	The number of males who have savings account	The number of females who have savings account
X	1200	900	300
Y	1000	600	400
Total	2200	1500	700

Current Account

Bank	The total number of people who have current account	The number of males who have current account	The number of females who have current account
X	800	500	300
Y	500	400	100
Total	1300	900	400

Answer: A

Quantity I

The total number of males who have account in bank X = $(900+500) = 1400$

The total number of females who have account in bank X = $(300+300) = 600$

So, the required difference = $(1400-600) = 800$

Quantity I = 800.

Quantity II

The number of females who have savings account in bank Y = 400

The number of females who have current account in bank Y = 100

So, the required sum = $(400+100) = 500$

Quantity II = 500.

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

20. Questions

Bank X

The total number of people who have account = 2000

The total number of people who have savings account = 60% of 2000 = 1200

The total number of people who have current account = 40% of 2000 = 800

Let the number of males who have savings account and females who have current account be 3m and m respectively.

The ratio of the number of females who have savings account and males who have current account = 3:5.
 Let it be 3n and 5n respectively.

Now,

$$3m + 3n = 1200$$

$$m + n = 400 \dots\dots\dots(I)$$

And,

$$m + 5n = 800 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$4n = 400$$

$$n = 100$$

Now, put the value of n in equation (I), we get,

$$m + n = 400$$

$$m + 100 = 400$$

$$m = 300$$

Therefore,

$$\text{The number of males who have savings account} = 3m = (3 \times 300) = 900$$

$$\text{The number of females who have savings account} = 3n = (3 \times 100) = 300$$

$$\text{The number of males who have current account} = 5n = (5 \times 100) = 500$$

$$\text{The number of females who have current account} = m = 300$$

Bank Y

$$\text{The total number of people who have savings account} = 1000$$

$$\text{The total number of people who have current account} = (1000/2) = 500$$

$$\text{The total number of people who have account} = (1000+500) = 1500$$

The ratio of the number of males who have savings account and females who have current account = 6:1.

Let it be $6a$ and a respectively. (Note: $16.67\% = 1/6$)

Also, the ratio of the number of females who have savings account and males who have current account = 1:1. Let it be b and b respectively.

Now,

$$6a + b = 1000 \dots\dots\dots(III)$$

And,

$$a + b = 500 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$6a + b = 1000$$

$$6a + 6b = 3000$$

$$5b = 2000$$

$$b = 400$$

Now, put the value of b in equation (III), we get,

$$6a + b = 1000$$

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$$6a + 400 = 1000$$

$$6a = 600$$

$$a = 100$$

Therefore,

The number of males who have savings account = $6a = (6 \times 100) = 600$

The number of females who have savings account = $b = 400$

The number of males who have current account = $b = 400$

The number of females who have current account = $a = 100$

Savings Account

Bank	The total number of people who have savings account	The number of males who have savings account	The number of females who have savings account
X	1200	900	300
Y	1000	600	400
Total	2200	1500	700

Current Account

Bank	The total number of people who have current account	The number of males who have current account	The number of females who have current account
X	800	500	300
Y	500	400	100
Total	1300	900	400

Answer: A

Bank X

The number of females who have current account = 300

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The new number of females who have savings account = $\{(300/3) \times 5\} = 500$

So, the new total number of females who have account (savings + current) = $(500 + 300) = 800$

And, the total number of females who have account (savings + current) originally = 600

Therefore, the required percentage increase = $[(800 - 600)/600] \times 100\% = \{(200/600) \times 100\}\% = 33.33\%$.

Hence, the correct answer is option A.